

Phase 1 (MVP) Subtask Deepening — epic → task → subtask (closes refinement R5)

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Volume 7 (Phase Execution), document 3 of 5. This spec **closes refinement item R5** (REFINEMENT_NOTES.md): Phase 0 (06-...) formalizes a full epic→task→subtask 3-tier breakdown, but Phase 1 (07-phase1-mvp-wbs.md) stops at epic→task. Here every Phase-1 task HVPN-P1-NNN is decomposed into PR-sized subtasks HVPN-P1-NNN.k (§11.4.93/.54 ids), each carrying a concrete **acceptance criterion** (falsifiable, captured-evidence per §11.4.5/.69/.107), the **§11.4.169 test types** that gate it, and an **estimated complexity** (XS/S/M/L T-shirt, sizing-only TARGET — never a date, §11.4.6). It is **spec-only**: it describes the subtasks and how "done" is proven; it does not build them. Every subtask traces to its parent task's Desc/Deliverable/Acceptance in 07-...; nothing is invented beyond decomposing the stated work. These .k rows feed the workable-items DB (workable-items-model.md §7 docs_chain context). Companion: dependency-graph.md (the DAG these subtasks inherit), subtask-deepening-p2.md, subtask-deepening-p3.md.

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0. Deepening conventions

Each parent task gets a subtask table: **id** (HVPN-P1-NNN.k, monotonic per parent, §11.4.54) · **Subtask** (≥ 6 -word title, §11.4.91) · **Acceptance** (falsifiable, captured-evidence) · **Tests** (§11.4.169 codes — UNIT INT E2E FA SEC CHAOS STRESS PERF BENCH SCALE UI UX REC CHAL, the 07-... §1 vocabulary) · **Cx** (complexity XS $\approx$ 0.5d, S $\approx$ 1-2d, M $\approx$ 2-3d, L $\approx$ 3-5d — sizing TARGET).

A subtask is complete only when its test types are green with a `test_diary` evidence path (`workable-items-model.md` §9). Subtask complexity sums to $\approx$ the parent task's 07-... Effort; it is a sizing aid, not a commitment (§11.4.6, 07-... §22). Phase-0 entry-gate items (HVPN-P1-001..006) are prerequisites tracked in 06-...; they are not re-deepened here.

1. E01 — Foundation (repo, proto, codegen)

HVPN-P1-010 — Monorepo + submodule layout (07-... §6; M)

id	Subtask	Acceptance	Tests	Cx
.1	Scaffold the snake_case component tree (helix-core/ helix-edge/ helix-go/ helix-proto/ helix-ui/ shims/ deploy/)	each dir present + decoupled (§11.4.28/.29); tree capture matches [SYNTHESIS §6]	UNIT,FA	S
.2	Wire .gitmodules + per-repo helix-deps.yaml (§11.4.31) for the six incorporated submodules	git submodule status lists containers/helix_qa/ challenges/docs_chain/ security/vision_engine	UNIT	XS
.3	upstreams/ recipes + install_upstreams per repo (§11.4.36)	git remote -v grep -c push = expected count on a fresh clone	FA	XS
.4	Assert no active CI (§11.4.156): workflows disabled/renamed	CM-NO-ACTIVE-CI green; git ls-files finds no .github/ workflows/*.yaml	FA,SEC	XS

HVPN-P1-011 — helix-proto agent protobuf + buf codegen (07-... §6; M)

id	Subtask	Acceptance	Tests	Cx
.1	Author agent.proto (Coordinator: Enroll/ WatchNetworkMap/ AdvertisePrefixes/ReportStatus) per [02 §4]	buf lint clean; service matches the [02 §4] contract	UNIT	S
.2	Wire buf generate → Go + Dart + Rust stubs	one source → three generated clients compile	UNIT,FA	S
.3	buf breaking gate on incompatible change	a removed field FAILs buf breaking (captured)	FA	XS
.4	Assert no hand-written client code exists (§8 [04_P1])	grep finds zero hand-rolled RPC client; generated- only	SEC	XS

HVPN-P1-012 — REST OpenAPI + Dart/TS client gen (07–... §6; S)

id	Subtask	Acceptance	Tests	Cx
.1	Define openapi/ helix.v1.yaml (enroll- tokens/devices/policies/ networks/stream)	OpenAPI 3.1 validates	UNIT	S
.2	Generate Dart + TS clients; round-trip a stub server	generated client round- trips; spec ^[?] handler drift FAILs the gen check	UNIT,FA	S

HVPN-P1-013 — Local dev infra via containers submodule (07–... §6; S)

id	Subtask	Acceptance	Tests	Cx
.1	internal/testinfra wrapper booting PG+Redis via pkg/boot/ pkg/compose/pkg/health (§11.4.76/.161 rootless)	integration tests boot infra on demand; no ad-hoc docker	INT,FA	S
.2	make test-infra-up/down with orphan-free teardown (§11.4.14)	teardown leaves zero orphan containers (captured podman ps)	INT,FA	XS

2. E02 — Store, RLS, no-log lint

Risk-first (§11.4.132): the irreversible correctness floor — ships before any feature reads/writes data.

HVPN-P1-020 — Migrations, enums, DB roles (07–... §7; M)

id	Subtask	Acceptance	Tests	Cx
.1	goose DDL: 11 tenant tables + enums per [02 §2.2]	migrations apply+rollback clean on fresh PG; \dt matches spec	UNIT,INT	M
.2	Three roles helix_owner/helix_app(non-super,non-bypass)/helix_sys	role grants verified; helix_app lacks rolsuper/rolbypassrls	INT,SEC	S
.3	Assert no connection/traffic table present (privacy floor)	\dt shows zero connections flows traffic packets table	SEC	XS

HVPN-P1-021 — sqlc typed queries + store.WithTenant (07–... §7; M)

id	Subtask	Acceptance	Tests	Cx
.1	sqlc-generated compile-time-checked queries	generated Queries compiles; sqlc vet clean	UNIT	S
.2	WithTenant(ctx,tid,fn) tx-scoped SET LOCAL app.tenant_id + WithSystem	a query forgetting WHERE tenant_id still returns only the active tenant's rows (INT)	UNIT,INT	M

HVPN-P1-022 — FORCE RLS + tenant_isolation policies (07–... §7; M)

id	Subtask	Acceptance	Tests	Cx
.1	ENABLE+FORCE RLS + tenant_isolation USING/WITH CHECK on all 10 tenant tables	every tenant table has the policy; \d+ capture	UNIT,INT	M
.2	store/rls_guard.go startup abort if helix_app can bypass RLS	startup aborts under a super role (captured exit)	UNIT,SEC	S
.3	Cross-tenant denial proof	a crafted tenant-A→tenant-B read returns ZERO rows (captured)	SEC	S
.4	Mid-tx drop rollback (no leak)	connection drop mid-tx rolls back; no partial cross-tenant write	CHAOS	S

HVPN-P1-023 — No-logging CI schema-lint + mutation (07–... §7; S)

id	Subtask	Acceptance	Tests	Cx
.1	tools/schemalint rejecting connection-log-shaped tables [02 §2.4]	green on the real schema;	UNIT,SEC	S

id	Subtask	Acceptance	Tests	Cx
		pre-build gate wired		
.2	Paired §1.1 mutation: inject connections(src,dst,bytes,ts) → lint FAILs	the planted table FAILs the lint (captured)	SEC	XS

3. E03 — Identity, enrollment, PKI

HVPN-P1-030 — Dual identity: OIDC + anonymous device tokens (07–... §8; M)

id	Subtask	Acceptance	Tests	Cx
.1	OIDC managed mode (Keycloak/Authentik) → tenant/role mapping	an OIDC user maps to a tenant/role (INT)	UNIT,INT	M
.2	Anonymous device enroll-tokens (no PII, Mullvad posture)	an anonymous device enrolls with zero PII persisted (SEC asserts DB/logs)	UNIT,SEC	S
.3	Token mint/store-hashed/consume-atomic	single-use, short-lived; concurrent double-consume rejected (SEC)	SEC,STRESS	S

HVPN-P1-031 — Enroll RPC (device key never leaves) (07–... §8; M · SLO2)

id	Subtask	Acceptance	Tests	Cx
.1	Token verify → IPAM-allocate → issue cert → persist device row (public WG key only)	device row holds only the WG public key (SEC asserts no private key)	UNIT,INT,SEC	M
.2	Emit device.enrolled; first NetworkMap path	event well-formed; map issued	INT,E2E	S
.3	SLO2 timing: enroll → first NetworkMap < 2 s	captured histogram p99 < 2 s	E2E,PERF	S

HVPN-P1-032 — PKI: device-cert issuance + tenant CA (07–... §8; M)

id	Subtask	Acceptance	Tests	Cx
.1	pki.Issue/Rotate/Revoke short-lived (24 h) mTLS certs vs per-tenant CA	a cert authenticates WatchNetworkMap (INT)	UNIT,INT	M
.2	CA-root secrecy (KMS/offline) + secret-leak audit (§11.4.10.A)	CA private key never in any app log (audit green)	SEC	S

HVPN-P1-033 — Cert rotation + revoke-<1 s (07-... §8; M · SLO3, AC6)

id	Subtask	Acceptance	Tests	Cx
.1	Auto-rotate before expiry over the existing channel	rotation at the boundary works (captured)	UNIT,INT	S
.2	device.revoke → drop WG peer from every map + edge enforce + cert revoked	revoked cert rejected immediately (SEC)	SEC	M
.3	SLO3 timing: revoke → edge enforcement < 1 s	captured revoke→enforcement timing p99 < 1 s	PERF	S

4. E04 — IPAM

HVPN-P1-040 — ULA /48 provisioning + AllocOverlayIP (07-... §9; M, D4)

id	Subtask	Acceptance	Tests	Cx
.1	Provision fd7a:helix:<rand>::/48 per tenant (: : 1=gateway) on tenant create	each tenant gets a distinct ULA /48 (INT)	UNIT,INT	S
.2	AllocOverlayIP deterministic + concurrency-safe	1000 concurrent allocations: no duplicate/gap (STRESS, captured)	UNIT,STRESS	M
.3	Restart-stable re-hydration of next_host	allocation resumes after restart with no reuse	INT	S

HVPN-P1-041 — 4via6 for colliding IPv4 LANs (07-... §9; M)

id	Subtask	Acceptance	Tests	Cx
.1	Stable site-id allocator per connector site	each site gets a stable id (INT)	UNIT,INT	S
.2	4via6 route synthesis → Peer.allowed_ips	one client reaches the same IPv4 CIDR via two connectors, disambiguated by site (E2E curl each)	E2E	M

5. E05 — Event backbone

HVPN-P1-050 — events.Bus Redis Streams + envelope (07-... §10; M, D3)

id	Subtask	Acceptance	Tests	Cx
.1	events.Bus interface + Redis Streams XADD/XReadGroup/XACK + §5.2 envelope	at-least-once delivery; idempotent reaction; replay harmless (INT)	UNIT,INT	M
.2	Consumer-crash re-delivery on restart	crash mid-handle re-delivers (CHAOS, captured)	CHAOS	S

HVPN-P1-051 — DLQ sweeper (XAUTOCLAIM) + metric (07–... §10; S)

id	Subtask	Acceptance	Tests	Cx
.1	XAUTOCLAIM reclaim + dead-letter routing + helix_events_dlq_total	poisoned event lands in DLQ after N retries; metric increments (captured)	UNIT,INT,CHAOS	S

HVPN-P1-052 — Event taxonomy producers (07–... §10; S)

id	Subtask	Acceptance	Tests	Cx
.1	Typed constructors for the [04_P1 §5.3] taxonomy (device., connector., route.conflict, policy.*, gateway.failover)	each producer emits a well-formed envelope on its trigger (INT capture)	UNIT,INT	S

6. E06 — Policy model & compiler

HVPN-P1-060 — Spec parse + group/host resolution (07–... §11; M)

id	Subtask	Acceptance	Tests	Cx
.1	Parse policies.spec jsonb (groups/hosts/acls/exitNodes)	valid spec parses; malformed rejected	UNIT	S
.2	Resolve group:*→device-ids + host→CIDRs vs advertised_prefixes	unknown group/host rejected; ambiguity emits route.conflict.detected	UNIT,INT	M
.3	Reject an ACL granting a revoked device	revoked-device grant rejected (SEC)	SEC	S

HVPN-P1-061 — Two-artifact compiler (AllowedIPs + verdict map) (07–... §11; L · AC2/AC3)

id	Subtask	Acceptance	Tests	Cx
.1			UNIT	M

id	Subtask	Acceptance	Tests	Cx
	Pure default-deny <code>compile(tenant, spec) → {visible, allowedIPs, exitNodes}</code>	golden-table tests pass; deterministic		
.2	Port-level verdict map (nftables/eBPF granularity)	a contractor granted 554,80 cannot reach port 22 (verdict-map SEC)	UNIT,SEC	M
.3	Determinism property: same spec $\Rightarrow$ byte-identical output	property test green over N runs (§11.4.50)	UNIT	S
.4	Perf: compile 1k-device tenant < 100 ms	captured timing < 100 ms	PERF	S

HVPN-P1-062 — Dry-run fail-closed validation + atomic activate/rollback (07–... §11; M · AC5)

id	Subtask	Acceptance	Tests	Cx
.1	Dry-run compile rejects unknown ref / uncovered dst / revoked grant	a bad policy never reaches the active version (fail-closed, captured)	UNIT,SEC	M
.2	Atomic version bump + <code>policy.compiled</code> emit + instant rollback	rollback restores the prior compiled set atomically (INT)	INT	S

7. E07 — Coordinator & WatchNetworkMap

Critical path; the < 1 s convergence promise lives here.

HVPN-P1-070 — In-memory per-tenant topology graph (07–... §12; M)

id	Subtask	Acceptance	Tests	Cx
.1	Hydrate per-tenant graph from PG on boot (no durable coordinator tables)	graph reconstructable from PG + events alone (INT)	UNIT,INT	M
.2	Event-driven incremental subgraph mutation	applying an event mutates only the affected subgraph (INT)	INT,SCALE	M

HVPN-P1-071 — `buildMap(node)` per-node map (07–... §12; M · AC2/AC3)

id	Subtask	Acceptance	Tests	Cx
.1	buildMap → NetworkMap{self,gateway,peers,dns,transport} with PeersVisibleTo filter	a node never appears in another's map unless policy grants it (SEC, captured)	UNIT,SEC	M
.2	relayEndpoint via gateway (MVP hub-and-spoke)	reachability matches the Phase-0 slice	UNIT	S

HVPN-P1-072 — WatchNetworkMap stream loop (07-... §12; L · AC5)

id	Subtask	Acceptance	Tests	Cx
.1	mTLS-auth → snapshot/ resume- from-known_version → delta+keepalive loop	reconnect with known_version=N resumes from deltas, no full resync (captured)	UNIT,INT,E2E	L
.2	Bounded per-stream send queue (slow-consumer shed)	a stalled consumer is shed without blocking others (CHAOS)	CHAOS	M

HVPN-P1-073 — Minimal-affected-set fan-out + reconcile metric (07-... §12; M · AC5/SLO1)

id	Subtask	Acceptance	Tests	Cx
.1	Compute minimal affected open- stream set; per-stream deltas (no full resync)	a 1-device change touches only the affected streams (INT count assertion)	UNIT,INT	M
.2	helix_reconcile_seconds histogram: event → delta-on-wire	SLO1 p99 < 1 s (captured histogram)	PERF	S

HVPN-P1-074 — 24 h soak / 10k-stream memory bound (07-... §12; M · SLO4)

id	Subtask	Acceptance	Tests	Cx
.1	10k simulated agents holding streams while flapping policies	SLO1 maintained under load (captured)	SCALE,STRESS	M
.2	24 h RSS time-series, bounded no-leak	coordinator memory bounded, no growth over 24 h (SLO4 graph)	PERF,SCALE	M

8. E08 — API surface + authz

HVPN-P1-080 — Connect handlers on Gin + mTLS resolver (07–... §13; M)

id	Subtask	Acceptance	Tests	Cx
.1	One multiplexed server: Connect (HTTP/2) alongside Gin REST	native agent + gRPC- Web caller both reach Coordinator	UNIT,INT	M
.2	authDevice(ctx) resolves identity from mTLS cert	an unauthenticated agent RPC is rejected (SEC)	SEC	S

HVPN-P1-081 — REST routes + OpenAPI + RBAC (07–... §13; M)

id	Subtask	Acceptance	Tests	Cx
.1	Gin routes (enroll-tokens/ devices/policies/networks/ audit)	routes match OpenAPI (INT)	UNIT,INT	M
.2	OIDC/session/API-token + RBAC (admin/operator/ member)	a member cannot mint enroll-tokens (403, SEC)	SEC	S
.3	RLS backstop (defense-in- depth)	RLS blocks cross-tenant read even if RBAC bypassed (INT)	INT,SEC	S

HVPN-P1-082 — GET /v1/stream WS/SSE fan-out (07–... §13; S)

id	Subtask	Acceptance	Tests	Cx
.1	Stream control- plane events to Console over WS/ SSE	a policy apply surfaces "policy v N active" to a connected Console within the convergence window (E2E)	UNIT,INT,E2E	S

HVPN-P1-083 — AdvertisePrefixes + ReportStatus RPCs (07–... §13; S)

id	Subtask	Acceptance	Tests	Cx
.1	AdvertisePrefixes accepts connector CIDRs → connector.prefixes.changed	advertised prefixes flow into the coordinator graph (INT)	UNIT,INT	S
.2	ReportStatus carries presence/ transport/rtt only (no bytes/flows)	SEC asserts the message has no byte/flow/ destination field	SEC	S

9. E09 — helix-core (Rust client/connector)

The shared client/connector/edge crate; durable evolution of the Phase-0 interfaces.

HVPN-P1-090 — Transport trait + plain-UDP + MASQUE (07–... §14; L · AC4, D1)

id	Subtask	Acceptance	Tests	Cx
.1	Harden the trait (send/recv/kind/effective_mtu, unreliable datagrams) into helix-transport	same trait drives client/connector/edge byte-for-byte	UNIT	M
.2	Production plain_udp impl	throughput baseline captured (BENCH)	UNIT,BENCH	S
.3	Production masque (quinn+h3+hand-rolled CONNECT-UDP/HTTP-Datagram, RFC 9298/9297/9221)	MASQUE flow classifies HTTP/3, no WG signature (tshark, SEC)	UNIT,SEC,E2E	L
.4	MASQUE goodput vs plain ≥ 50%	captured BENCH ≥ 50% of plain	BENCH	S

HVPN-P1-091 — helix-wg boringtun wrapper (07–... §14; M)

id	Subtask	Acceptance	Tests	Cx
.1	Wrap Tunn (handshake/encrypt/decrypt/4 verdicts)	in-process handshake + data round-trip identical plaintext	UNIT	M
.2	Sustained 1 GB transfer + 2-min rekey	no handshake stall (STRESS); rekey at boundary works (captured)	E2E,STRESS	M

HVPN-P1-092 — Orchestrator + network-map reconciler (07–... §14; L · AC5)

id	Subtask	Acceptance	Tests	Cx
.1	Three-loop orchestrator (tun→wg→transport, transport→wg→tun, timer)	loops drive a plain-UDP slice (E2E)	UNIT,E2E	M
.2	Reconciler consuming streamed MapUpdate (snapshot/delta) without restart	a delta adds/removes a peer with no reconnect of unrelated state (FA)	FA	M
.3	Mid-stream coordinator-restart re-sync from known_version	re-syncs cleanly (CHAOS)	CHAOS	S

HVPN-P1-093 — Auto-escalation ladder (07–... §14; M · AC4)

id	Subtask	Acceptance	Tests	Cx
.1	Walk TransportPolicy.order	with plain WG blocked, client escalates to	UNIT,E2E,SEC	M

id	Subtask	Acceptance	Tests	Cx
	(plain-udp→lwo→masque-h3) on handshake failure, honor allow_user_override	MASQUE and stays up (E2E transport=="masque-h3")		
.2	Deterministic ladder across N runs (§11.4.50)	identical escalation outcome over N=3	UNIT	S

HVPN-P1-094 — Kill-switch + DNS-leak protection (07–... §14; L · AC7)

id	Subtask	Acceptance	Tests	Cx
.1	Core state machine drives platform firewall on Down/Reconnecting	host-side pcap during a forced drop shows ZERO leaked packets (SEC)	UNIT,SEC	L
.2	DNS pinned to overlay resolver while up; no plaintext DNS on drop	zero plaintext DNS leaked (SEC capture)	SEC	M
.3	State survives an abrupt interface flap	kill-switch holds across flap (CHAOS)	CHAOS	S

HVPN-P1-095 — helix-ffi (flutter_rust_bridge v2) surface (07–... §14; M)

id	Subtask	Acceptance	Tests	Cx
.1	start(ClientConfig)/stop() + codegen	Dart drives connect/disconnect; codegen clean	UNIT,FA	S
.2	status_stream(StreamSink<TunnelStatus>) mirroring the broadcast channel	Dart renders live status from the stream alone (UI golden + FA ×3)	UNIT,UI,FA	M

10. E10 — helix-edge (data plane)

Language fixed by Phase-0 G4 (default lean Rust, shares helix-transport).

HVPN-P1-100 — MASQUE termination on :443/udp + masquerade (07–... §15; L · AC4)

id	Subtask	Acceptance	Tests	Cx
.1			UNIT,E2E	L

id	Subtask	Acceptance	Tests	Cx
	Terminate MASQUE/H3, extract HTTP Datagrams → WG → kernel WG	a valid client tunnels through (E2E, captured)		
.2	Native decoy site for non-CONNECT-UDP probes	a scanner at :443 sees a plain website (SEC)	SEC	M
.3	CPU/Gbps within the G4 budget	captured BENCH within budget	BENCH	S

HVPN-P1-101 — Kernel-WG peer programming + relay (07–... §15; M · AC2)

id	Subtask	Acceptance	Tests	Cx
.1	Program kernel WG peers from the coordinator map	peers appear in wg show (INT)	UNIT,INT	M
.2	Hub-and-spoke relay peer[?]peer via gateway	client reaches an authorized connector LAN host through the relay (E2E curl)	E2E	M

HVPN-P1-102 — Edge verdict-map enforcement + revoke drop (07–... §15; M · AC3/AC6, SLO3)

id	Subtask	Acceptance	Tests	Cx
.1	Apply the compiler's port-level verdict map (nftables/eBPF)	an unauthorized port is dropped (AC3 SEC, captured)	UNIT,INT,SEC	M
.2	Drop a revoked device's sessions on device.revoked	revoke removes the kernel peer < 1 s (SLO3, captured); negligible p99 add (PERF)	PERF,SEC	S

HVPN-P1-103 — Edge hardening posture (07–... §15; S)

id	Subtask	Acceptance	Tests	Cx
.1	Rootless Podman, read-only rootfs, seccomp, NET_ADMIN-only, no SSH	container runs unprivileged except NET_ADMIN (SEC)	SEC	S
.2	Fail-static: killing control plane doesn't drop data-plane tunnels	established tunnels survive a control-plane kill (CHAOS capture)	CHAOS	S

11. E11 — helix-ui (Flutter)

All UI under OpenDesign tokens (§11.4.162) — light+dark, no ad-hoc CSS.

HVPN-P1-110 — helix_design system (OpenDesign tokens, Material 3) (07–... §16; M)

id	Subtask	Acceptance	Tests	Cx
.1	Brand-tokenized palette + signature components (ConnectButton/StatusChip/ExitPicker/ShieldIndicator/AdaptiveScaffold), light+dark	visual-regression goldens pass for every component in both themes (UI)	UNIT,UI	M
.2	No overlapping/overlaid labels (§11.4.162 layout invariant)	layout-audit golden clean; palette reacts to connection state (REC vision-verified §11.4.159)	UI,REC	S

HVPN-P1-111 — Shared shell `runHelixApp(flavor,...)` + Riverpod binding (07-... §16; M)

id	Subtask	Acceptance	Tests	Cx
.1	One tree → three flavors (Access/Connector/Console)	same widget tree renders all three flavors (UI)	UNIT,UI	M
.2	Riverpod providers = pure function of the FFI status stream	UI state deterministic over a replayed status stream (FA)	FA	S

HVPN-P1-112 — Access app screens (enroll → connect → reach) (07-... §16; L · AC9)

id	Subtask	Acceptance	Tests	Cx
.1	Enroll (paste/scan token) → connect → ExitPicker/authorized-network list	golden tests for each screen (UI)	UI	M
.2	Real enroll→connect→reach-LAN journey window-scoped MP4 (§11.4.143/.159)	vision verdict confirms the authorized host loaded (UX/REC)	UX,E2E,REC	M

HVPN-P1-113 — Console (web) admin screens (07-... §16; L · AC5/AC9)

id	Subtask	Acceptance	Tests	Cx
.1	Devices/enroll-tokens/policy editor (validate→activate→rollback) + live /v1/stream feed	a policy edit reflects on devices < 1 s with a live "policy v N active" feed (E2E/REC, AC5)	UI,E2E,REC	L
.2	RBAC hides admin actions from member	member sees no admin action (SEC)	SEC	S

HVPN-P1-114 — Connector daemon UI/headless control (07-... §16; M · AC9)

id	Subtask	Acceptance	Tests	Cx
.1	Enroll + advertise CIDRs + show prefixes/reachability; headless daemon mode	a connector advertises a LAN; its prefixes appear in Console + a client's map (E2E)	UI,E2E,REC	M

12. E12 — Platform tunnel shims

MVP platforms only: iOS, Android, Linux. The shim is the only platform-specific code.

HVPN-P1-120 — Linux tun shim (Rust-native) (07-... §17; S · AC1/AC2)

id	Subtask	Acceptance	Tests	Cx
.1	Linux core drives the TUN directly (reference for the netns rig)	the Access Linux build runs the full enroll→connect→reach journey on the netns rig (E2E)	E2E,FA	S

HVPN-P1-121 — Android VpnService + JNI shim (07-... §17; M · AC7/AC9)

id	Subtask	Acceptance	Tests	Cx
.1	VpnService owns the TUN; JNI loads helix-core.so	tunnel up on a real phone (device REC)	E2E,UX,REC	M
.2	Kill-switch → VpnService always-on/lockdown	kill-switch blocks plaintext on drop (SEC capture)	SEC	S

HVPN-P1-122 — iOS NEPacketTunnelProvider shim (within G3 budget) (07-... §17; L · AC7/AC9)

id	Subtask	Acceptance	Tests	Cx
.1	NEPacketTunnelProvider configures settings + pumps packetFlow [?] Rust core	tunnel up on a real device (device REC)	E2E,UX,REC	L
.2	Honor the G3 memory ceiling over a sustained transfer	captured Instruments RSS within ceiling (PERF)	PERF	M
.3	Kill-switch + DNS-leak hold	zero plaintext/DNS leak on drop (SEC)	SEC	S

13. E13 — Deploy

HVPN-P1-130 — helixvpctl bootstrap + ops CLI (Cobra) (07–... §18; M · AC1)

id	Subtask	Acceptance	Tests	Cx
.1	init generates tenant/admin/CA/overlay-/48/creds/gateway-keys/quadlets	helixvpctl init on a fresh host produces a runnable deploy (FA, captured)	UNIT,FA	M
.2	gateway keys/enroll-token/policy apply/device revoke subcommands	policy apply ./policy.jsonc validates + activates (E2E)	E2E	S

HVPN-P1-131 — Podman quadlets (rootless, systemd-managed) (07–... §18; M · AC1)

id	Subtask	Acceptance	Tests	Cx
.1	Render helixd/helix-edge/postgres/redis quadlets, rootless, read-only rootfs	systemctl --user start helixvpn-pod brings the stack up rootless (FA)	INT,FA	M
.2	helixd connects to PG as the non-superuser role (RLS enforced)	the app role cannot bypass RLS (SEC)	SEC	S

HVPN-P1-132 — On-demand integration infra via containers submodule (07–... §18; S)

id	Subtask	Acceptance	Tests	Cx
.1	QA harness boots+tears down the full stack via the submodule (§11.4.76)	the E2E suite boots+tears down the stack itself (FA, captured)	INT,FA	S

HVPN-P1-133 — Compose dev-loop + K8s Phase-2 shape (reference) (07–... §18; S)

id	Subtask	Acceptance	Tests	Cx
.1	Docker Compose dev-loop (local) + reference stateless-coordinator K8s manifest	dev compose boots PG+Redis+helixd; the K8s manifest is lint-valid (not deployed at MVP)	INT	S

14. E14 — QA, SLOs, DoD certification

Anti-bluff capstone: the §11.4.40 full-suite retest, §11.4.107 recorded evidence, §11.4.27 100%-type coverage.

HVPN-P1-140 — netns + DPI + netem E2E rig (CI-ready) (07–... §19; M)

id	Subtask	Acceptance	Tests	Cx
.1	Promote the Phase-0 rig fed by the real control plane (lanA + nft WG-block + tc netem)	make e2e runs the full slice unattended	E2E,FA	M
.2	Re-runnable -count=3 identically (§11.4.98)	three identical runs (captured)	FA	S

HVPN-P1-141 — SLO instrumentation + dashboards (07–... §19; S · SLO1-4)

id	Subtask	Acceptance	Tests	Cx
.1	Prometheus metrics + Grafana-as-code + alert rules at the §2 SLO targets	all four SLOs measured + asserted in CI thresholds (captured histograms, alert-on-breach)	PERF,FA	S

HVPN-P1-142 — Security test suite (07–... §19; L · AC3/AC6/AC7/AC8)

id	Subtask	Acceptance	Tests	Cx
.1	RLS cross-tenant denial + key-never-leaves + mTLS-reject + kill-switch/DNS pcap + revoke timing + secret-leak audit (§11.4.10.A)	every assertion green with captured evidence	SEC,SCALE	L
.2	Paired §1.1 mutations (drop an RLS policy / weaken kill-switch) FAIL their gate	each mutation FAILs (captured)	SEC,CHAOS	M

HVPN-P1-143 — Stress + chaos suite (§11.4.85) (07–... §19; M)

id	Subtask	Acceptance	Tests	Cx
.1	Sustained load + concurrency + boundary across the control plane	no deadlock/leak under 10× concurrency	STRESS,SCALE	M
.2	Failure injection (Redis kill / PG drop / coordinator SIGKILL / disk-full)	every fault recovers to a consistent state with a captured recovery_trace.log	CHAOS	M

HVPN-P1-144 — HelixQA Challenge bank + vision-verified recordings (07–... §19; M · AC1-AC9)

id	Subtask	Acceptance	Tests	Cx
.1	challenges/helix_qa bank dispatching real user journeys, PASS only on captured evidence	the 8-AC journey set scores PASS with a vision-confirmed MP4 per AC (no frozen/bluff frame §11.4.107)	CHAL,REC,UX	M
.2	Analyzer self-validated (golden-good/bad); broken build FAILs	a deliberately broken build scores FAIL	CHAL	S

HVPN-P1-145 — Full-suite retest + DoD certification gate (07–... §19; M · AC1-AC9/SLO1-4)

id	Subtask	Acceptance	Tests	Cx
.1	§11.4.40 release-gate sweep on a clean baseline (pre/post-build + on-device + meta-mutation + Challenge + no-log lint + CONTINUATION sync)	all of §21 green simultaneously on a from-zero install (§11.4.108 fresh deploy)	FA,E2E,CHAL	M

15. E15 — Governance & release

HVPN-P1-150 — Workable-items DB projection + docs-chain sync (07–... §20; M)

id	Subtask	Acceptance	Tests	Cx
.1	cmd/workable-items/loader: parse leaves → upsert items/test_diary (git-tracked DB §11.4.95)	md[?]db byte-identical round-trip (§11.4.93)	UNIT,INT,FA	M
.2	.docs_chain/contexts/wbs.yaml registered (§11.4.106)	a leaf edit re-syncs exports out-of-the-box; verify is the CI gate	INT,FA	S

HVPN-P1-151 — Release tagging + multi-upstream publish (07–... §20; S · AC1-AC9)

id	Subtask	Acceptance	Tests	Cx
.1	Cut <HELIX_RELEASE_PREFIX>-1.0.0-mvp (§11.4.151) on main + every owned submodule	git tag -l '<PREFIX>-*' enumerates the whole release across repos	FA	S
.2	Publish to all upstreams via merge-onto-latest-main (no force-push §11.4.113)	the tag is reachable on every remote; HVPN-P1-145 cert	FA	S

id	Subtask	Acceptance	Tests	Cx
		report is the tag's evidence		

16. Subtask roll-up + coverage note

Epic	Parent tasks	Deepened subtasks	Notes
E01 Foundation	4	12	repo/proto/codegen/dev-infra
E02 Store+RLS+lint	4	11	risk-first correctness floor
E03 Identity/PKI	4	11	revoke-<1s + key-never-leaves
E04 IPAM	2	5	ULA /48 + 4via6 (D4)
E05 Events	3	4	Redis Streams (D3)
E06 Policy compiler	3	9	two-artifact, default-deny
E07 Coordinator	5	11	<1s convergence + 24h soak
E08 API+authz	4	8	Connect+REST+RBAC+RLS backstop
E09 helix-core	6	16	Transport/WG/orchestrator/kill-switch/FFI
E10 helix-edge	4	9	MASQUE term + verdict-map + fail-static
E11 helix-ui	5	10	OpenDesign + 3 flavors
E12 shims	3	6	Linux/Android/iOS
E13 deploy	4	7	helixvpnctl + quadlets
E14 QA/DoD	6	11	rig/SLO/security/chaos/Challenge/cert
E15 governance	2	4	items-DB + release tag
Total	59	134	—

Subtask complexity sums to $\approx$ the parent task effort (07–... §22, ~273 person-days excl. Phase-0 prereqs); it is a **sizing aid, not a commitment** (§11.4.6). Every subtask is anti-bluff: its acceptance is a captured-evidence assertion (§11.4.5/.69/.107), metadata-only PASS is forbidden (§11.4.1), and it is complete only with a test_diary evidence path (workable-items-model.md §9). The .k ids are monotonic per parent and never renumbered (§11.4.54). The full set of these subtasks feeds the §11.4.93 DB via the subtask-deepening-p1.md docs_chain source (workable-items-model.md §7).

Sources verified

- 07-phase1-mvp-wbs.md §6–§20 (every task Desc/Deliverable/Acceptance/Effort/Tests), §21 traceability, §22 effort roll-up — read 2026-06-26.
- 06-phase0-spike-wbs.md §0.1 (the 3-tier breakdown R5 aligns to) — read 2026-06-26.
- REFINEMENT_NOTES.md R5 (subtask-tier asymmetry to close) — read 2026-06-26.
- Sibling workable-items-model.md (§3 field mapping, §6 test diary, §9 complete-definition), dependency-graph.md (the inherited DAG) — authored this volume.
- Constitution anchors
§11.4.5/.6/.40/.54/.58/.69/.85/.91/.93/.98/.107/.108/.113/.132/.143/.151/.156/.159/.162/.169 — read 2026-06-26.

Honest boundary (§11.4.6): subtasks decompose the *stated* parent-task work; no new scope is invented. All complexity figures are sizing TARGETs, never date commitments; device-bound subtasks (iOS HVPN-P1-122., *Android HVPN-P1-121.*) carry the §11.4.3 SKIP-with-reason fallback where no device is reachable.